

Seminar Report

Security Analysis of Ephemeral Kubernetes in HPC Environments

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Abstract

While Ephemeral Kubernetes offers rapid, sub-90-second cluster deployments for High-Performance Computing (HPC) environments, its original architecture inherently prioritizes speed over security. By relying on shared Network File System (NFS) storage for cluster bootstrapping, the system exposes critical cryptographic materials, most notably full cluster-admin credentials, creating severe risks of persistent credential exposure and privilege escalation. This paper analyzes these fundamental security flaws and proposes two architectural enhancements to secure the cluster lifecycle. First, we introduce a Key-Based Credential Distribution system that eliminates persistent storage risks through authenticated, transient file delivery. Second, we implement a Token-Based Access Control model that enforces the principle of least privilege by replacing unrestricted admin credentials with scoped join tokens. Together, these methodologies ensure that Ephemeral Kubernetes can satisfy the rigorous security demands of multi-tenant HPC facilities without sacrificing its signature agility and performance.

Declaration on the use of ChatGPT and comparable tools in the context of examinations

In this work I have used ChatGPT or another AI as follows:

- ☐ Not at all
- ☒ During brainstorming
- ☐ When creating the outline
- ☐ To write individual passages, altogether to the extent of 0% of the entire text
- ☐ For the development of software source texts
- ☒ For optimizing or restructuring software source texts
- ☒ For proofreading or optimizing
- ☐ Further, namely: -

I hereby declare that I have stated all uses completely.

Missing or incorrect information will be considered as an attempt to cheat.

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- 2 Kube.config will be saved in a secure server folder on the cluster manager, which is accessible using Key(separate for each node, provided through overlay) authentication. Figure created using <https://app.diagrams.net/> . . . 6

List of Abbreviations

API	Application Programming Interface
CA	Certificate Authority
CN	Common Name
GPU	Graphics Processing Unit
HPC	High-Performance Computing
HTTP	Hypertext Transfer Protocol
KSI	Kind Slurm Integration
MPI	Message Passing Interface
NFS	Network File System
PKI	Public Key Infrastructure
SHA	Secure Hash Algorithm
VM	Virtual Machine
WW	Warewulf

1 Introduction

While Kubernetes is not as popular in production HPC environments as it is in cloud VMs due to complexity in adaptability, maintainability, performance bottlenecks, and security, several genuine approaches attempt to fill this gap.

Zhou et al. [5] use Torque-Operator to bridge Kubernetes and HPC clusters managed by TORQUE, enabling container orchestration while keeping the HPC environment intact. However, this requires persistent external components outside Slurm jobs, adding infrastructure complexity and security concerns in multi-user environments. Chazapis et al. [2] run Slurm under Kubernetes, converting user services into Slurm scripts that execute as Singularity containers on HPC compute nodes. This approach does not support Kubernetes ClusterIP services, lacks GPU mapping and port forwarding, and relies on static Slurm allocations, limiting scheduling flexibility. Decker et al. [3] introduce KSI (Kind Slurm Integration), which deploys complete Kubernetes clusters within Slurm jobs, providing full Kubernetes features. However, KSI only supports single-node deployments, lacks multi-node MPI-style parallel jobs, and suffers from poor network performance.

Ephemeral Kubernetes[1] addresses challenges in adaptability, maintainability and performance bottlenecks by integrating container orchestration directly into existing HPC stacks via Warewulf with minimal architectural changes. This framework provides native Kubernetes performance and services, and could also be used to dynamically negotiate HPC cluster capacity between on-demand and batch clusters[4]. The current Ephemeral Kubernetes implementation prioritises deployment speed and operational simplicity over security, relying on shared Network File System (NFS) storage to distribute sensitive cryptographic materials. Specifically, the Kubernetes cluster-administrator configuration file (`kube.config`) containing full cluster-admin credentials is stored on NFS accessible to all cluster nodes. This architectural decision creates severe security vulnerabilities:

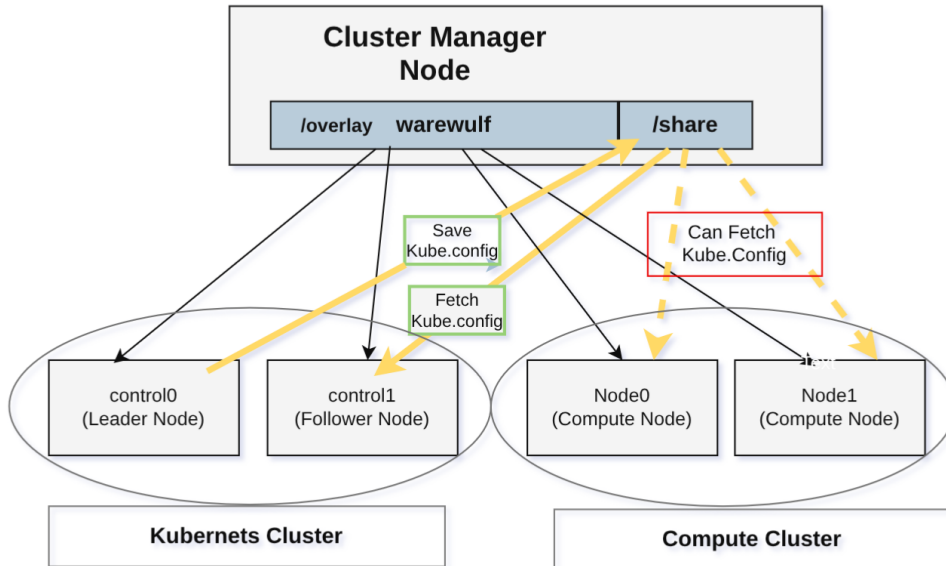


Figure 1: Unsecured ephemeral Kubernetes setup: The (`kube.config`) (used to join the cluster) is directly accessible through the `/share` folder; anyone can use it and deploy privileged pods and get root access on the node. Figure created using <https://app.diagrams.net/>

- **Credential Exposure:** Any compromised node or malicious container with NFS access can steal cluster-admin credentials.
- **Privilege Escalation:** Stolen credentials enable deployment of privileged containers, leading to root access on any cluster node.
- **Persistent Risk:** Credentials remain exposed indefinitely on NFS throughout the cluster lifecycle.
- **No Revocation:** Once credentials are copied, they cannot be revoked or expired.
- **Lateral Movement:** Compromising a single worker node provides a path to compromise the entire cluster.

These vulnerabilities are unacceptable for production HPC environments, particularly those serving multiple tenants, processing sensitive data, or subject to compliance Requirements.

2 Background and Security Challenges

2.1 Ephemeral Kubernetes Architecture Overview

Ephemeral Kubernetes is built upon Warewulf (WW), a stateless node provisioning system commonly deployed in HPC environments. The architecture consists of three primary components: a Warewulf host server that provisions operating system images, control plane nodes that manage the Kubernetes cluster, and worker nodes that execute containerised workloads.

Unlike traditional Kubernetes deployments, where nodes maintain persistent state, Ephemeral Kubernetes nodes boot from network-provisioned images and lose all local state upon reboot. The system leverages two distinct Warewulf images:

- A **control image** containing Kubernetes control plane components [6] (`kube-apiserver`, `kube-scheduler`, `kube-controller-manager`, `etcd`), HAProxy, and Keepalived for high availability.
- A **base image** for worker nodes containing only the `kubelet` and container runtime.

Both images include pre-loaded container images to enable offline deployment without internet connectivity.

A critical architectural component is the shared Network File System (NFS) storage mounted at `/share` on all nodes. This shared storage serves multiple purposes: synchronising configuration between nodes, storing pre-loaded container images, coordinating leader election during cluster bootstrap, and distributing cryptographic credentials required for cluster formation.

2.2 Cluster Formation and Node Join Process

The cluster formation process in Ephemeral Kubernetes follows a coordinated sequence that relies heavily on the shared NFS storage for synchronisation.

2.2.1 Leader Election & Cluster Initialization

When multiple control plane nodes boot simultaneously, a leader election mechanism determines which node initialises the cluster. The leader node creates a `/share/leader` folder on the NFS share. Other control nodes see that the `/share/leader` folder already exists, recognise themselves as followers and wait for cluster initialisation to complete.

The elected leader node performs several security-critical operations:

- **Certificate Generation:** The leader runs `kubeadm init`, which generates the cluster's Public Key Infrastructure (PKI) [7], including the Certificate Authority (CA) certificates, API server certificates, and etcd certificates.
- **Credential Creation:** Kubeadm generates the administrator configuration file (`kube.config`) [8] containing client certificates signed by the cluster CA. These certificates include the Common Name (CN) `kubernetes-admin` and Organisation (O) `system:masters`, granting unrestricted cluster access.
- **Credential Distribution:** The leader copies all generated certificates and the admin configuration file to the NFS share: `/share/pki/ca.crt`, `/share/pki/ca.key`, `/share/pki/etcd/ca.crt`, `/share/pki/etcd/ca.key`, and `/share/kube.config`.
- **Ready Signal:** After completing initialisation, the leader creates the `/share/leader_ready` file to signal other nodes.

2.2.2 Follower and Worker Node Join Process

Non-leader nodes poll for the `leader_ready` file every 5 seconds. Once detected, nodes retrieve credentials from NFS and join the cluster:

2.2.3 Follower and Worker Node Join Process

Non-leader nodes poll for the `leader_ready` file every 5 seconds. Once detected, nodes retrieve credentials from NFS and join the cluster. All joining nodes authenticate with the API server using the client certificate embedded in the `kube.config` file, which is validated against the cluster CA (`ca.crt`).

- **Control Plane Follower Nodes:** Copy CA certificates from `/share/pki/` to `/etc/kubernetes/pki/` as they need to run control plane components such as the API server, etcd, and scheduler. They then start the Phylactery service for HAProxy configuration management and join as control plane members using `kubeadm join -discovery-file /root/.kube/config -control-plane -v=5`.
- **Worker Nodes:** Do not require any PKI certificates as they only run workloads and do not host any control plane components. They simply read the credentials from `/share/kube.config` and join the cluster using `kubeadm join -discovery-file /root/.kube/config -v=5`.

Both node types use the same administrator credentials for cluster joining.

2.3 Node Rejoin and Recovery Mechanism

A distinguishing feature of Ephemeral Kubernetes is its handling of node reboots. Since nodes are stateless, every time a node goes down, it wipes its local state, so a rebooted node appears as a new entity attempting to join an existing cluster. However, the cluster's internal state retains records of the node from its previous session, causing hostname and IP conflicts that prevent rejoining. The Phylactery service addresses this by performing cleanup operations to remove these stale memberships before the node attempts to join again. This recovery mechanism explicitly requires the (`kube.config`) file to be persistently available on the NFS share. Without the administrator-level credentials provided by (`kube.config`), the service cannot authenticate with the API to purge the previous entries from the etcd database and the Kubernetes node list.

3 Security Vulnerabilities in the Ephemeral Model

As mentioned in the Node Rejoin section, all the nodes in the Kubernetes cluster plane need to have persistent access to cluster config files and certificates, which exist on `/share` folder on the cluster manager, which can give admin-level access to the Kubernetes cluster to any user on any node.

- **Privilege Escalation:** Stolen credentials enable deployment of privileged containers [9], leading to root access on any cluster node. A user with admin access to a Kubernetes cluster can escalate this to root access on any node of the cluster, for example, by deploying a privileged container that mounts `/etc/shadow` and replaces the root password.
- **No Revocation Mechanism:** Once credentials are copied, they cannot be revoked or expired. If an attacker copies the `kube.config` file, there is no way to invalidate those credentials without recreating the entire cluster.
- **Lateral Movement:** Compromising a single worker node provides a path to compromise the entire cluster. The shared credential model means that breaching any single node's security gives an attacker the keys to the entire infrastructure.

4 PKI Certificates and Their Usage

Kubernetes uses certificates and keys to secure the cluster and verify identities. Each certificate serves a specific purpose in the cluster's security model.

- **Cluster CA (`ca.crt/ca.key`):** The master certificate for the entire cluster. The `ca.crt` file lets nodes verify they're talking to the real API server. The `ca.key` file is used to create certificates for all cluster components. This is the most critical key - anyone with `ca.key` can create admin credentials and take over the cluster.

- **Service Account Keys** (`sa.key/sa.pub`): These keys create and verify tokens for pods running in the cluster. When a pod needs to talk to the API server, it gets a token signed with `sa.key`. The API server checks this token using `sa.pub`; all control plane nodes must use identical keys so tokens work no matter which API server receives the request.
- **Front Proxy CA** (`front-proxy-ca.crt/front-proxy-ca.key`): Used for API aggregation, which lets additional API servers connect to the main Kubernetes API. All control plane nodes must share these certificates for extension APIs to work correctly.
- **etcd CA** (`etcd/ca.crt/etcd/ca.key`): Protects communication between etcd database instances. When a new control plane node joins, it uses `etcd/ca.key` to create certificates that let it communicate securely with existing members.
- **Admin Kubeconfig File**: The `kube.config` file contains the cluster CA certificate, a client certificate proving admin identity, and a private key. Possession of this file grants complete administrative control over the cluster.

The current setup stores `ca.key` and `Kube.config` in the shared `/share` directory, which creates a security risk. Anyone accessing this folder can use `ca.key` or `Kube.config` to generate admin certificates and gain full cluster access. A better approach would use secured folder for storing these sensitive files, which is accessible only with some authentication, or share only limited-permission join tokens and service account tokens that only allow specific operations like node management.

5 Proposed Security Enhancement: Key-Based Credential Distribution

To address the critical security vulnerabilities inherent in the NFS-based credential distribution model, we propose a Key-based authentication system that eliminates persistent credential exposure while maintaining the operational benefits of ephemeral cluster deployment.

5.1 Architecture Overview

The enhanced architecture replaces the shared NFS storage of sensitive credentials with a secure file server running on the Warewulf cluster manager. This server implements a Key-based authentication mechanism where each node receives a unique Key that grants access to the credentials and config file.

The secure file server operates as a Flask-based HTTP service listening on the cluster manager's internal network interface (typically `10.0.0.3:8000`). Unlike the previous NFS-based approach, where credentials remained indefinitely accessible at `/share/pki/` and

`/share/kube.config`, the new system stores all sensitive materials in a private directory (`/var/lib/warewulf/private-secrets`) with strict filesystem permissions (mode 700), preventing access by any user or process except root, and the server running on the cluster manager will serve those files to anyone with a valid key.

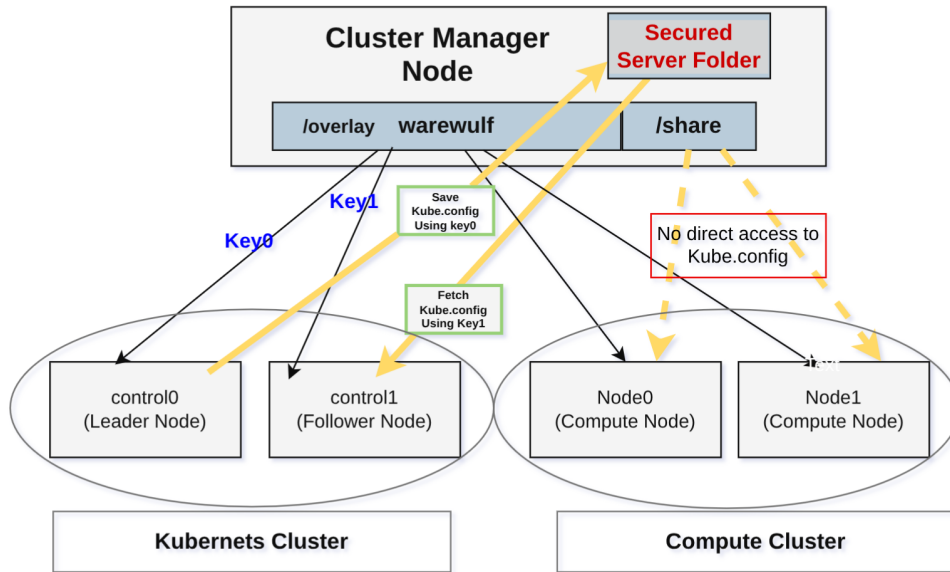


Figure 2: Kube.config will be saved in a secure server folder on the cluster manager, which is accessible using Key(separate for each node, provided through overlay) authentication. Figure created using <https://app.diagrams.net/>

5.2 Key Generation and Distribution

Prior to cluster deployment, the cluster administrator executes a Key generation script (`generate-Keys.sh`) that creates unique SHA-256 Keys for each node registered in Warewulf. Each Key is stored as a JSON file in `/var/lib/warewulf/Keys/` containing metadata including the associated node name, creation timestamp, and expiration status.

The Keys are distributed to nodes through Warewulf's overlay system. During node provisioning, each node receives its unique Key via a node-specific file (`/etc/k8s-Key.{NODE_NAME}`) injected into the boot image. This approach ensures Keys are delivered securely during the provisioning process without network transmission or shared storage exposure.

5.3 Secure Credential Upload Mechanism

The leader node, after completing `kubeadm init` and generating the cluster PKI, uploads sensitive materials to the secure file server rather than copying them to NFS storage. The upload process uses HTTP POST requests with binary payloads:

```
curl -X POST --data-binary @/etc/kubernetes/kube.config \
    "http://10.0.0.3:8000/upload/kube.config/$Key"
```

The server validates the Key before accepting the upload and sanitizes filenames to prevent directory traversal attacks. Only two files are uploaded: the cluster administrator configuration (`kube.config`) and a compressed archive of CA certificates (`pki.tar.gz`) containing exclusively:

- Cluster CA certificate and Key (`ca.crt`, `ca.Key`)
- Service account Keys (`sa.Key`, `sa.pub`)

- Front proxy CA certificate and Key (`front-proxy-ca.crt`, `front-proxy-ca.Key`)
- etcd CA certificate and Key (`etcd/ca.crt`, `etcd/ca.Key`)

Critically, the archive excludes node-specific certificates (API server certificates, etcd peer certificates) that should be uniquely generated per node, preventing certificate reuse vulnerabilities.

5.4 Key-Authenticated Download and Self-Destruction

Follower nodes and worker nodes retrieve credentials by presenting their unique Keys to the secure file server:

```
curl -f -s "http://10.0.0.3:8000/config/$Key" \
    -o /root/.kube/config
curl -f -s "http://10.0.0.3:8000/certs/$Key" \
    -o /tmp/pki.tar.gz
```

The server validates the Key's existence and ensures it has not been marked as expired. Upon successful credential download and cluster join completion, the node immediately deletes its Key from local storage, removing the file `/etc/k8s-Key.{NODE_NAME}`. This Key self-destruction ensures that once a node successfully joins the cluster, the authentication credential no longer persists in the node's memory or filesystem, eliminating the risk of Key theft from a running node.

Here is the shortened version:

“`latex`

5.5 Key Lifecycle and Node Rejoin Mechanism

The Key lifecycle accommodates the ephemeral nature of Warewulf-provisioned nodes. When a node reboots, it loses all state including its Key. However, Warewulf re-provisions the node from the base image during each boot, delivering the same Key through the overlay system. The node uses this Key to re-authenticate, download credentials, rejoin the cluster (with Phylactery handling cleanup of the previous node instance), and subsequently deletes the Key. This cycle repeats indefinitely across reboots without accumulating credential exposure. For operational deployments, administrators should regenerate Keys periodically using `generate-Keys.sh` and rebuild Warewulf overlays with `wwctl overlay build` to ensure previously compromised Keys cannot be reused.

5.6 Security Improvements

The Key-based system delivers several critical security enhancements over the NFS-based approach:

- **Elimination of Persistent Exposure:** Sensitive credentials exist only in a private directory on the cluster manager after cluster formation, no longer stored indefinitely on a network-accessible filesystem.
- **Node-Specific Authentication:** Each node authenticates using a unique Key tied to its identity, enabling audit logging of credential access for forensic investigation.

- **Reduced Lateral Movement Risk:** Compromising a worker node no longer grants access to cluster-admin credentials as the Key is deleted immediately after joining.
- **Time-Bounded Attack Window:** The attack window for Key theft is limited to the brief period between node boot and successful cluster join, typically under 90 seconds, drastically reducing the temporal attack surface compared to indefinite NFS exposure.

5.7 Implementation Considerations

The secure file server is deployed as a systemd service (`ww-secure-server.service`) that starts automatically with the cluster manager, logging all access attempts to `/var/log/ww-file-server.log` for security monitoring. The installation script (`k8s-install.sh`) detects the security mode via the `USE_SECURE_MODE` environment variable, switching nodes to Key-authenticated downloads instead of direct NFS access. The Key is deleted immediately after `kubeadm join` completes successfully, eliminating residual credential risk. Since the node filesystem is entirely RAM-based (`tmpfs`), a reboot would naturally clear the Key regardless, but explicit deletion provides defense-in-depth. For rejoin scenarios, Warewulf automatically delivers the same Key through the node overlay, enabling seamless rejoining without manual intervention. To enforce credential rotation, administrators must manually regenerate Keys and rebuild overlays before nodes reboot. “

6 Alternative Approach: Token-Based Access Control

Instead of sharing the `kube.config` file and sensitive certificate keys in the `/share` directory, a more secure approach uses limited-permission tokens and carefully selected certificates that follow the principle of least privilege.

6.1 Recommended Files for Shared Storage

To secure the cluster while maintaining the resilience required for HPC environments, this architecture shares only the specific credentials necessary for node lifecycle operations. Unlike the original configuration which exposed full administrative control, these files follow the principle of least privilege.

6.1.1 Bootstrap Join Token (`join-command.txt`)

This file contains the `kubeadm join` command equipped with a bootstrap token [10]. This token is required for all nodes (workers and control plane) to join the cluster.

- **Scope:** The token’s permissions are strictly limited to the node joining process.
- **Security:** It cannot be used to access cluster resources, read secrets, or perform administrative operations. Even if a malicious actor obtains this token, they cannot query the API server or modify the cluster state.

6.1.2 Phylactery Service Account Token (`phylactery-token.txt`)

This token allows the Phylactery service to clean up stale node entries from the Kubernetes node list managed by the API server and from the etcd member database upon reboot and rejoin (unlike the original approach, where the service relied on the unrestricted `kube.config` to clear these entries). It carries minimal permissions restricted to node management (list, delete, drain), ensuring it cannot access cluster secrets or perform broader administrative tasks.

- **Scope:** Strictly limited to node lifecycle management operations only. The token can list, drain, and delete node objects from the Kubernetes node list but has no permissions beyond these specific operations.
- **Security:** Even if a malicious actor obtains this token, they cannot access cluster secrets, deploy or modify workloads, generate new credentials, or perform any administrative operations. The token provides no path to privilege escalation unlike the original `kube.config` which granted full cluster-admin access.

6.1.3 Certificate Key (`certificate-key.txt`)

This key is required exclusively for control plane follower nodes to fetch the certificates and private keys (`ca.key`, `etcd/ca.key`, `sa.key`, `front-proxy-ca.key`, `ca.crt`, `etcd/ca.crt`, `etcd/server.crt`, `etcd/server.key`) necessary for running control plane services such as the API server, etcd, and scheduler. In the original architecture, these were stored in plain text on NFS, directly accessible to any node or user with share access. The token-based approach eliminates this by having the leader node encrypt all these sensitive files into a secure bundle during `kubeadm init`. The `certificate-key.txt` acts as the decryption password for this bundle. When a control plane follower joins or rejoins after a reboot, it presents this key to the API server, receives the encrypted bundle, and decrypts it locally to obtain the certificates it needs, ensuring raw private keys never reside on NFS at any point.

- **Scope:** Strictly limited to control plane nodes only. Worker nodes do not require this key as they never need to obtain or decrypt any certificate bundle.
- **Security:** Even if a malicious actor obtains this key, they must still actively interact with the API server to retrieve the encrypted bundle, generating auditable logs. Critically, raw private keys such as `ca.key` are never directly accessible on NFS, eliminating the primary attack vector of the original architecture.

6.2 Cluster Formation in Token-Based Architecture

Similar to the original architecture, a leader election mechanism determines which control plane node initialises the cluster. Once elected, the leader runs `kubeadm init` to generate the cluster PKI and bootstrap the control plane. Unlike the original architecture where all sensitive certificates and `kube.config` were copied directly to the `/share` folder, the leader instead places only three limited-access credentials on NFS:

- `join-command.txt`: Allows worker and control plane follower nodes to join the cluster with no privileges beyond node registration.

- **phylactery-token.txt**: Allows the Phylactery service to clean up stale node entries from the Kubernetes node list and etcd database upon node reboot and rejoin.
- **certificate-key.txt**: Allows control plane follower nodes to fetch the encrypted bundle of sensitive certificates and private keys directly from the API server, ensuring raw keys never reside on NFS.

Once these files are placed on NFS and the `leader_ready` signal is created, follower and worker nodes join the cluster using only these minimal credentials, each serving a single specific purpose in the cluster lifecycle.

7 Discussion

This study analyzed the security posture of Ephemeral Kubernetes in HPC environments, identifying a critical vulnerability in its reliance on shared NFS storage for cluster bootstrapping. While the original architecture prioritises deployment speed and operational simplicity, achieving sub-90-second boot times, it does so by violating fundamental security principles. The centralised exposure of the `kube.config` file and the Cluster CA key (`ca.key`) creates a single point of failure where the compromise of any worker node or the shared file system results in the total compromise of the Kubernetes control plane. This security model is unsuitable for multi-tenant HPC centres where workload isolation is paramount.

To mitigate these risks, we proposed two distinct architectural improvements. The first approach, **Key-Based Credential Distribution**, addresses the *transport and storage* vector. By replacing the passive NFS share with an active, authenticated file server, we eliminate the persistent availability of credentials. The use of unique, node-specific keys distributed via Warewulf overlays ensures that authentication is tied to hardware identity, while the "self-destruct" mechanism (deleting keys post-join) significantly narrows the temporal attack window. This approach transforms the credential distribution from a static, globally accessible resource into a transient, audited transaction.

The second approach, **Token-Based Access Control**, addresses the *privilege scope* vector. By adhering to the principle of least privilege, this method replaces the omnipotent administrator credentials with scoped Bootstrap Tokens and Service Account tokens. Even if these tokens are intercepted, an attacker's capabilities are strictly limited to node registration or specific cleanup tasks, preventing them from accessing secrets, escalating privileges, or dominating the cluster.

While both approaches independently improve security, they operate at different layers of the infrastructure. The Key-based system secures the *delivery pipeline*, protecting the credentials during transit and storage. The Token-based system secures the *payload*, ensuring that the credentials themselves grant minimal access. For production HPC environments requiring defence-in-depth, the optimal configuration would likely combine these methodologies: using the secure Key-based server to deliver the restricted Tokens. This combination would ensure that credentials are not only difficult to intercept but also hold limited value if compromised, thereby enabling Ephemeral Kubernetes to meet the rigorous security standards of modern supercomputing facilities without sacrificing its dynamic capabilities.

8 Conclusion

Ephemeral Kubernetes offers a promising solution for integrating dynamic container orchestration into traditional HPC environments, though its original design [1] left significant security challenges unaddressed. As noted in the foundational work, the reliance on unsecured NFS storage for credential distribution creates a single point of failure where the compromise of any node can lead to total cluster takeover. This report has presented concrete architectural solutions to these previously identified vulnerabilities. We introduced a Key-based delivery system to secure the transport and storage of credentials, alongside a Token-based access control model that minimizes the privileges granted to joining nodes. By implementing these proposed enhancements, HPC centers can effectively close the security gaps inherent in the original Ephemeral Kubernetes model, enabling safe, multi-tenant operation while retaining the rapid provisioning capabilities that make the platform unique.

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A Work sharing

If you worked in a group, describe here how you distributed the work and the actual contributions of each peer.

A.1 Hans

...

A.2 Peter

...

B Code samples

This is part of the appendix...